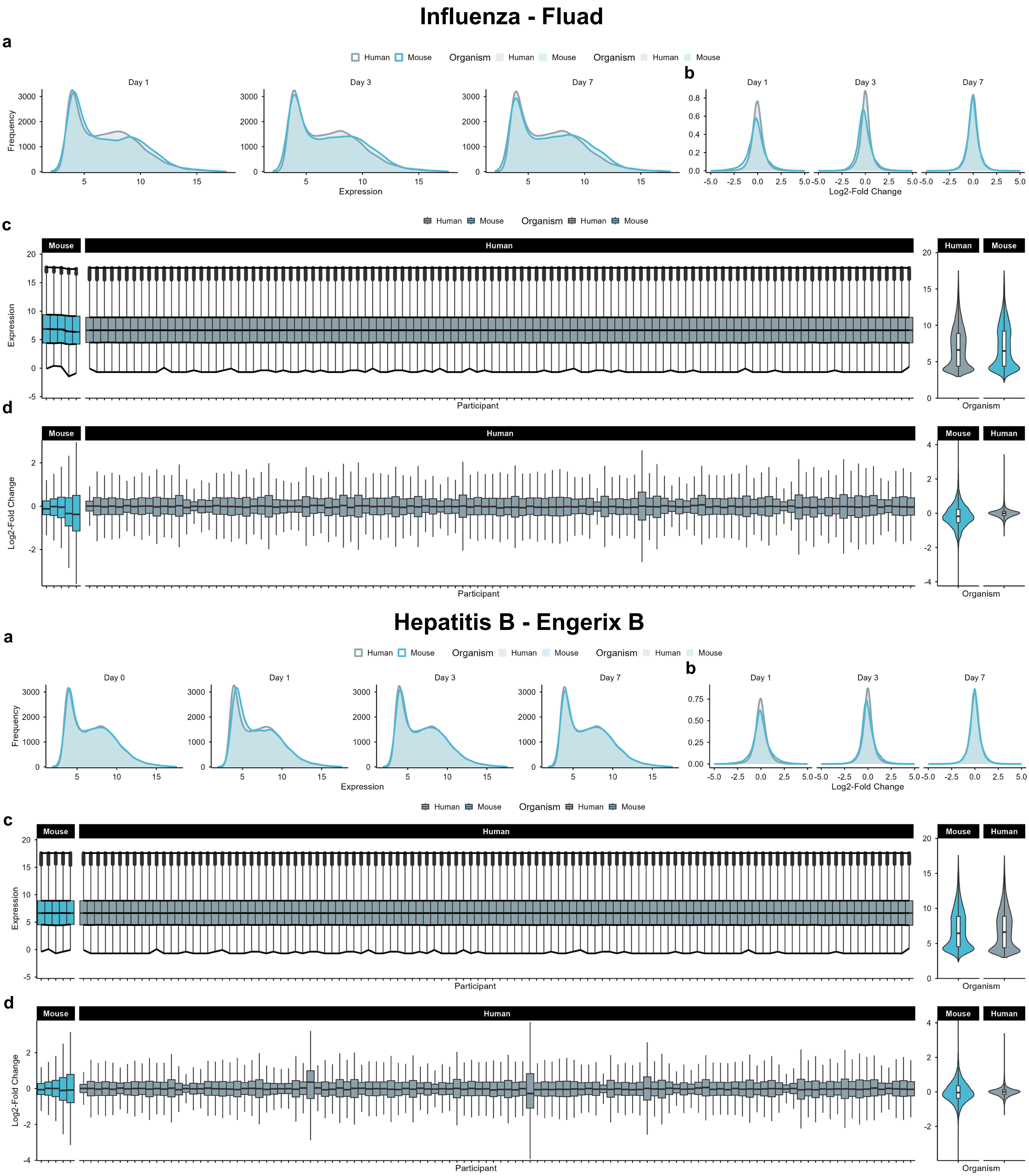
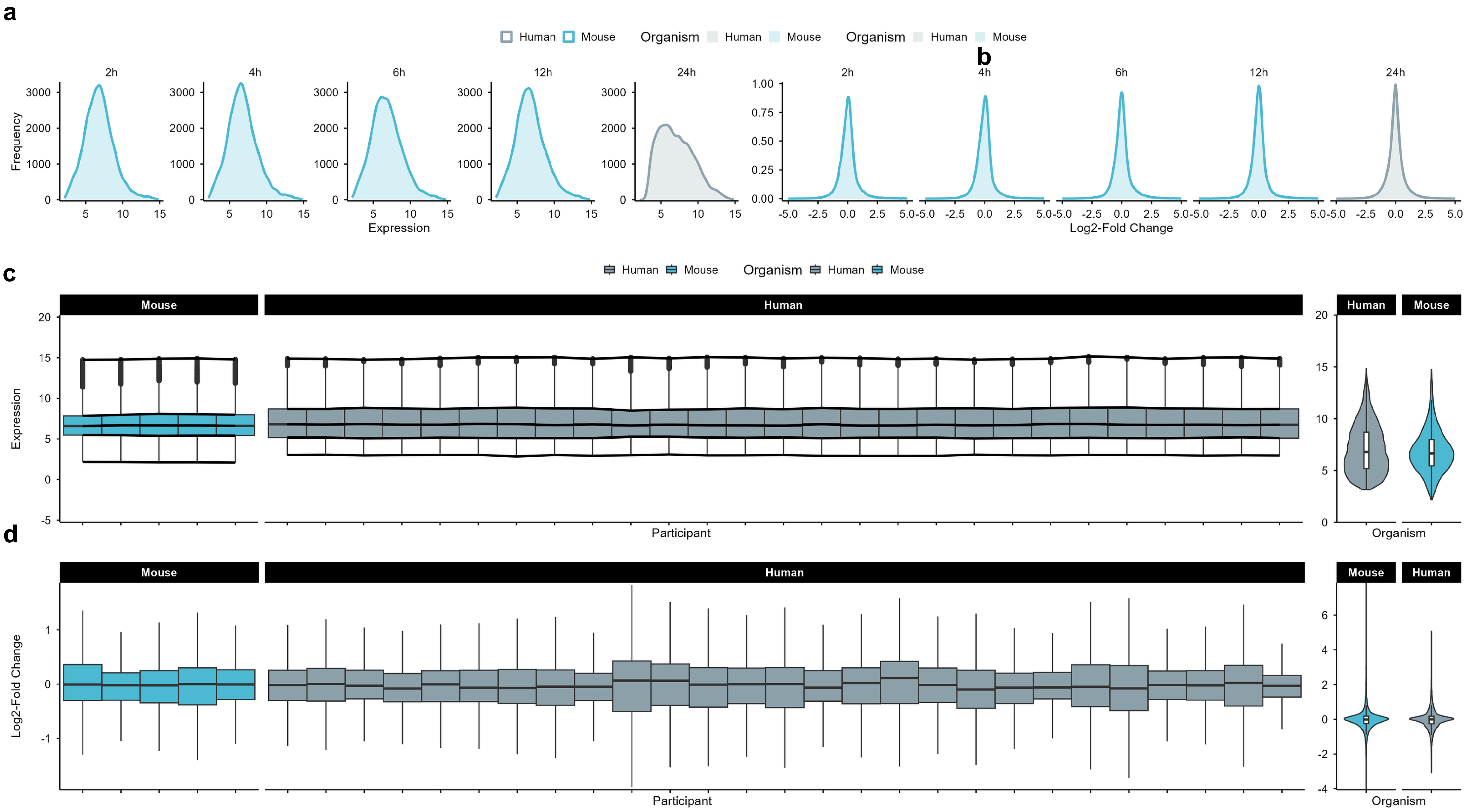
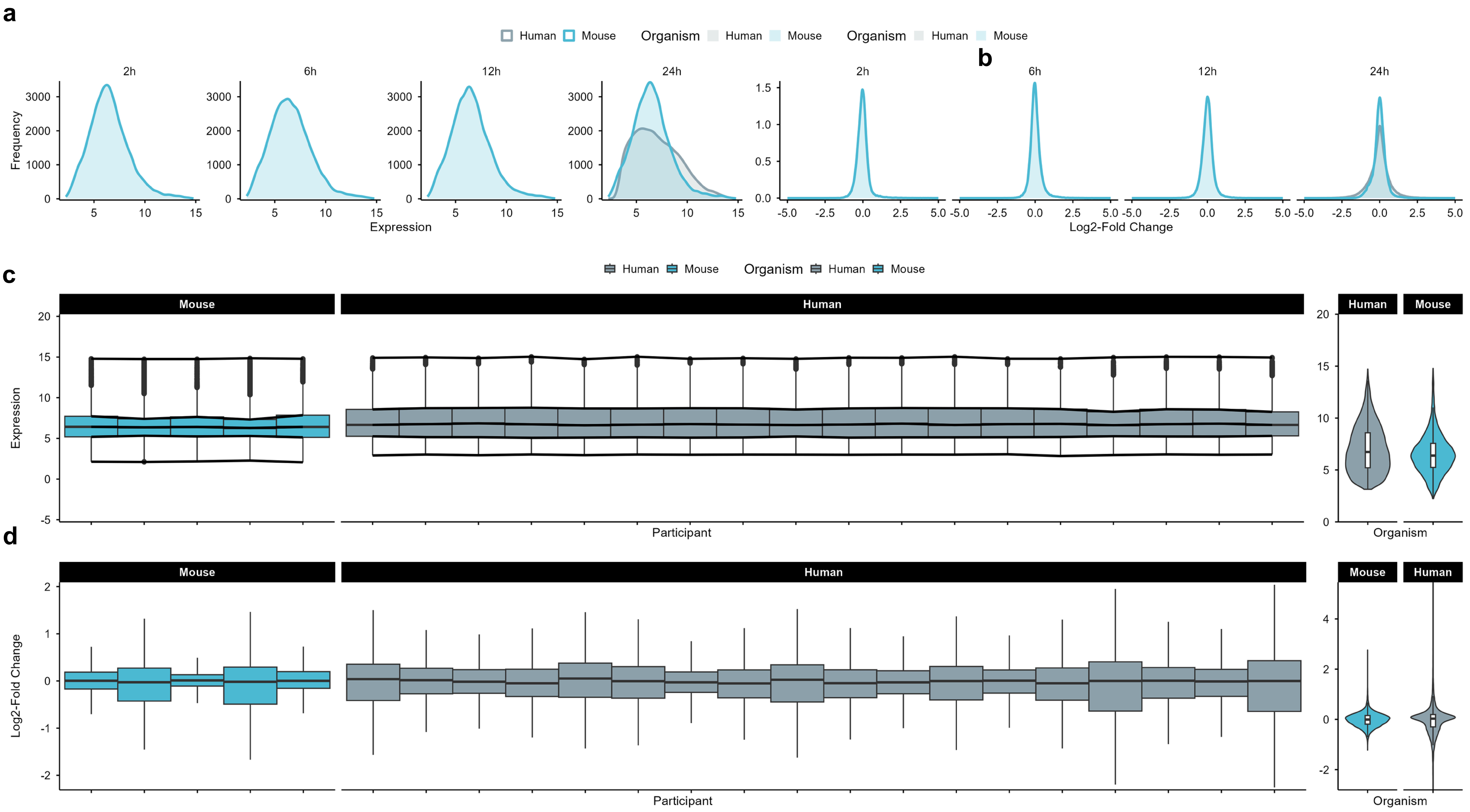


Assessment of gene expression data quality of vaccination against influenza and hepatitis b. Density plots of (a, e) log2 expression and (b, f) log2-fold change distributions, respectively, in humans (gray) and mice (blue). Boxplots of log2 values (c, g) and log2-fold changes (d, h) for each participant on day 1.

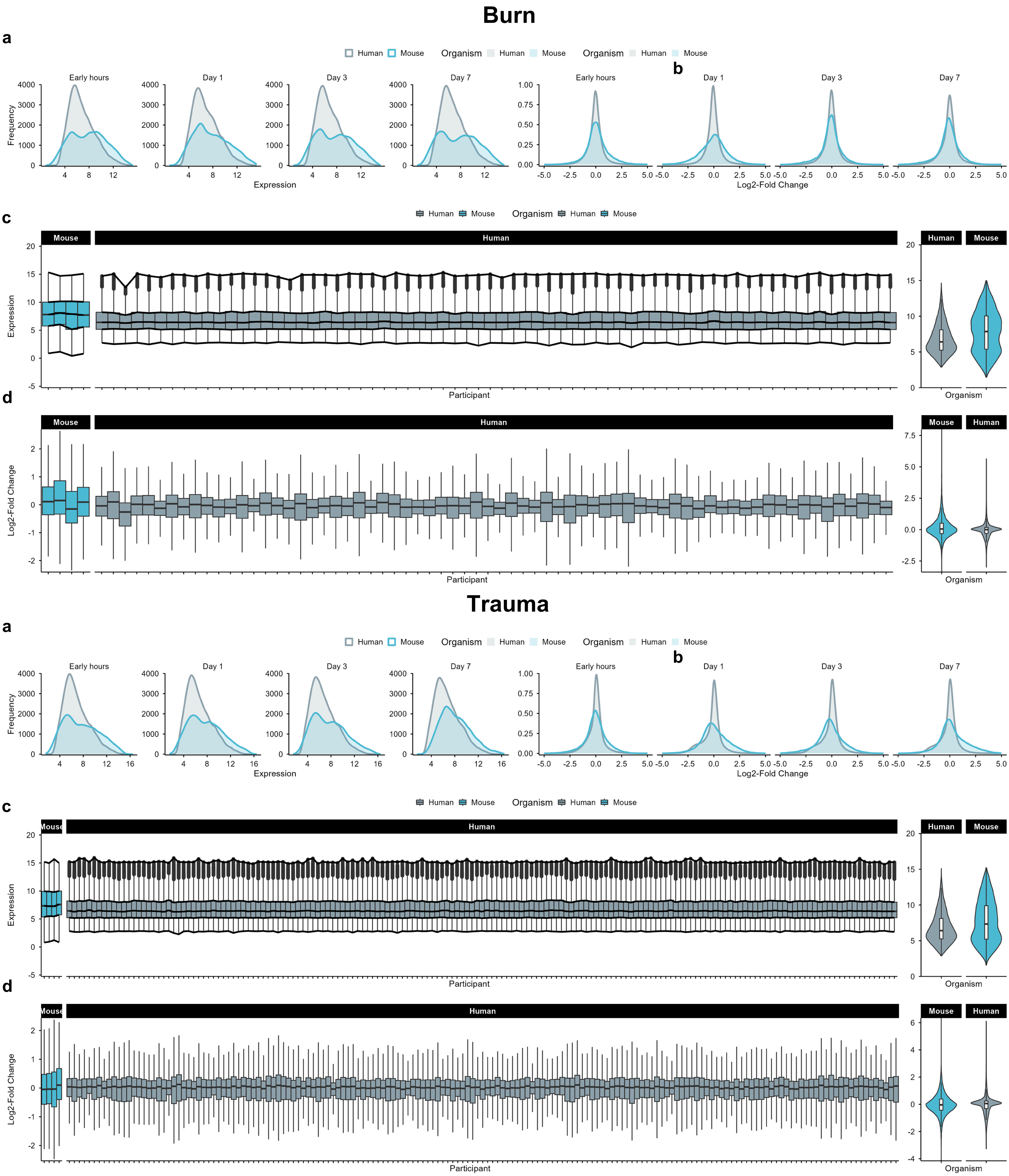
Staphylococcus aureus



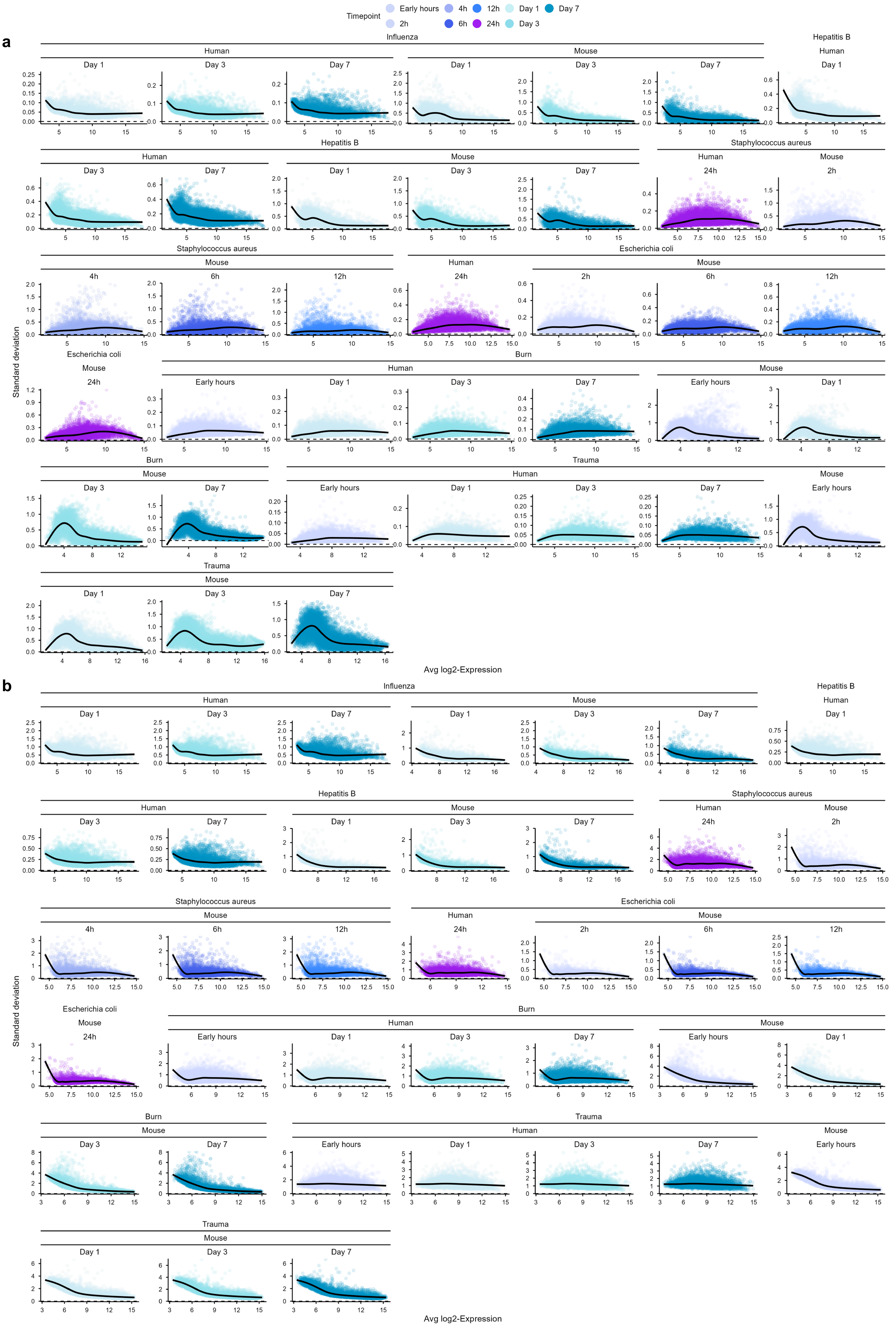
Escherichia coli



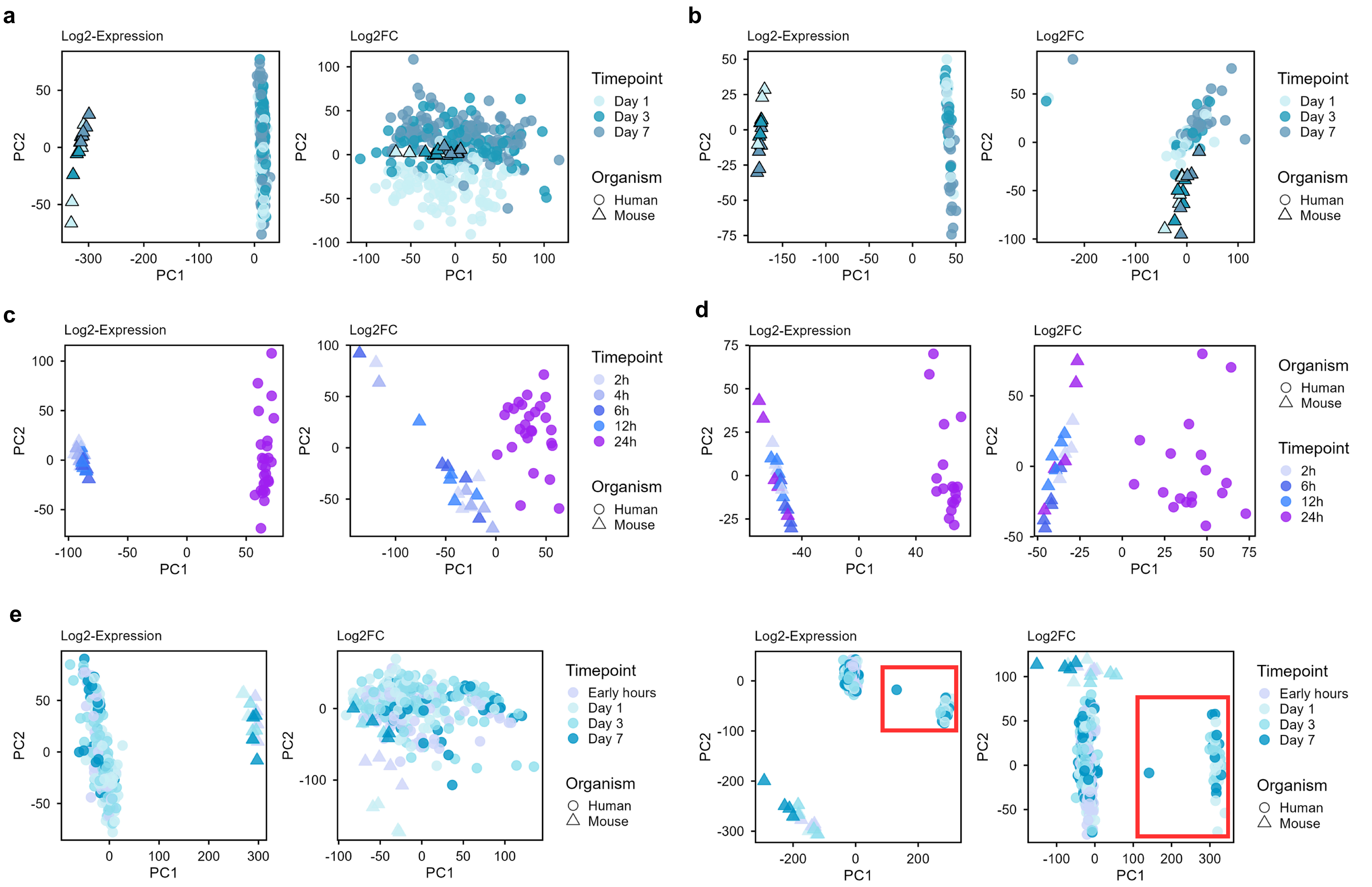
Assessment of gene expression data quality of infection models of S. aureus and E. coli. Density plots of (a, e) log2 expression and (b, f) log2-fold change distributions, respectively, in humans (gray) and mice (blue). Boxplots of log2 values (c, g) and log2-fold changes (d, h) for each participant on day 1.



Assessment of gene expression data quality of injury models of burn and trauma-hemorrhage. Density plots of (a, e) log2 expression and (b, f) log2-fold change distributions, respectively, in humans (gray) and mice (blue). Boxplots of log2 values (c, g) and log2-fold changes (d, h) for each participant on day 1.

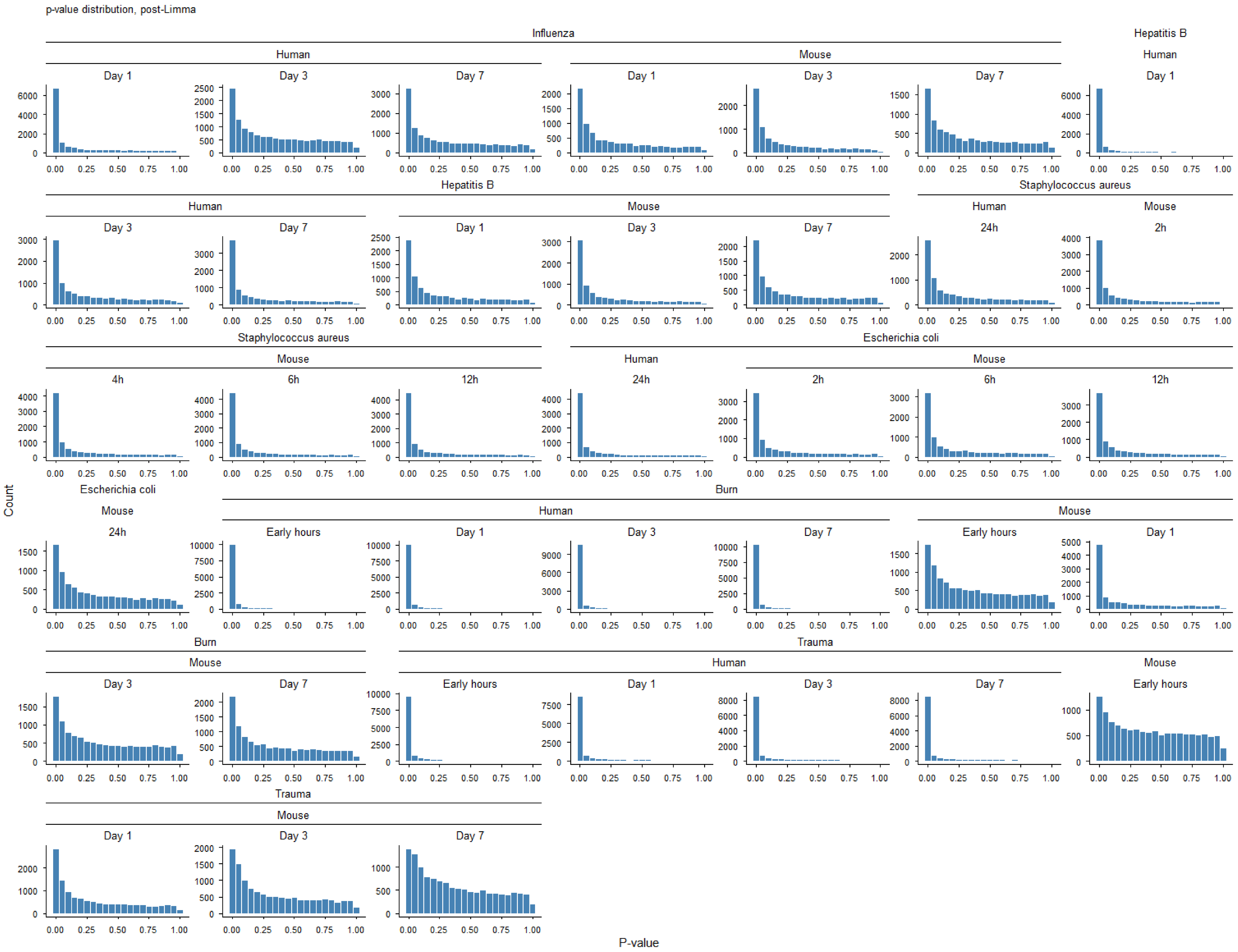


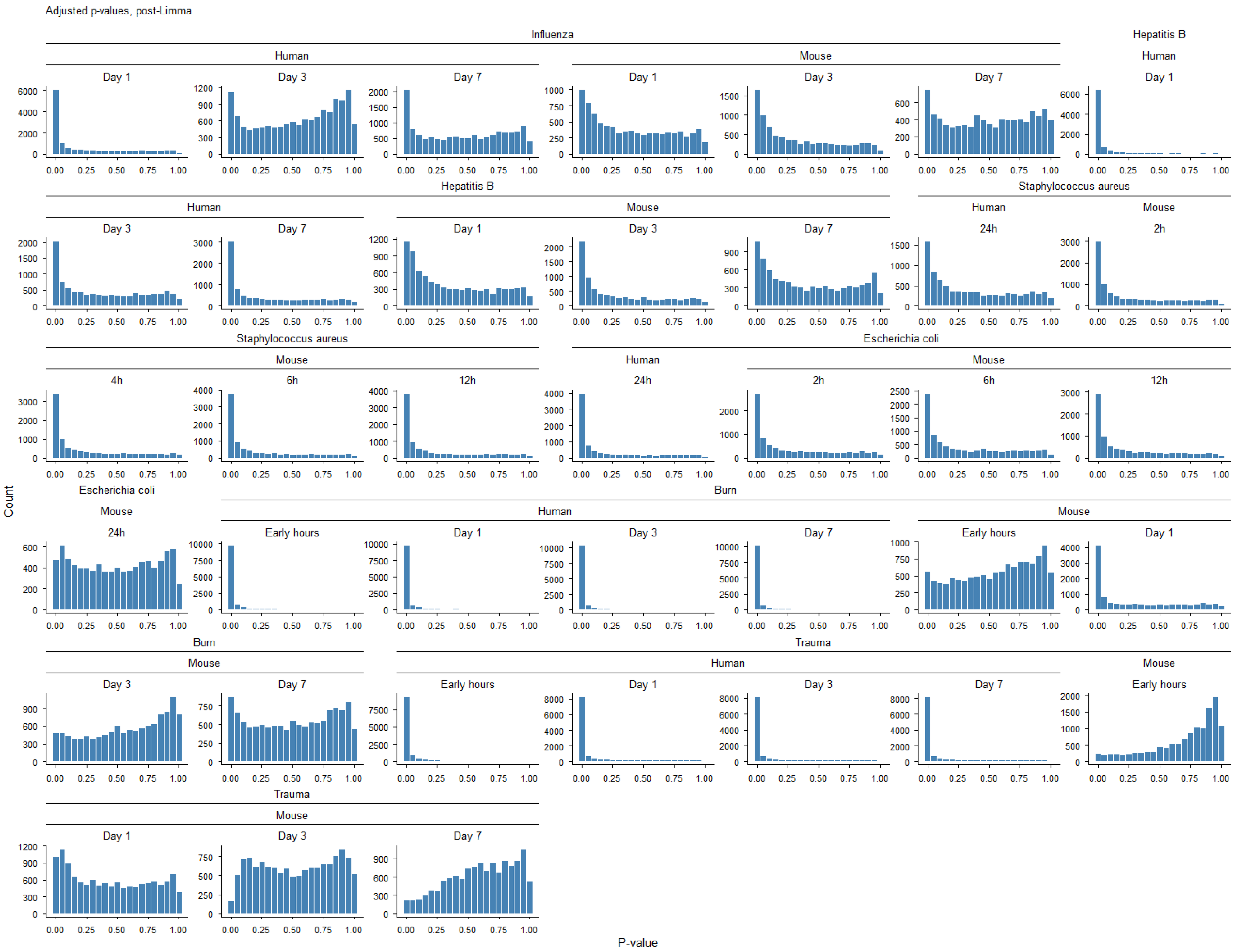
Evaluation of the mean-variance relationship (SA plots) in gene expression data before (a) and after (b) linear model processing. The plots display the residual standard deviation (sigma) as a function of the average log2-expression for each gene across multiple human and mouse datasets under various conditions. The filtering of low expressed genes and the application of the eBayes function with trend = TRUE and robust = TRUE parameters resulted in a significantly more stable and horizontal trend line (smoothed black curve).



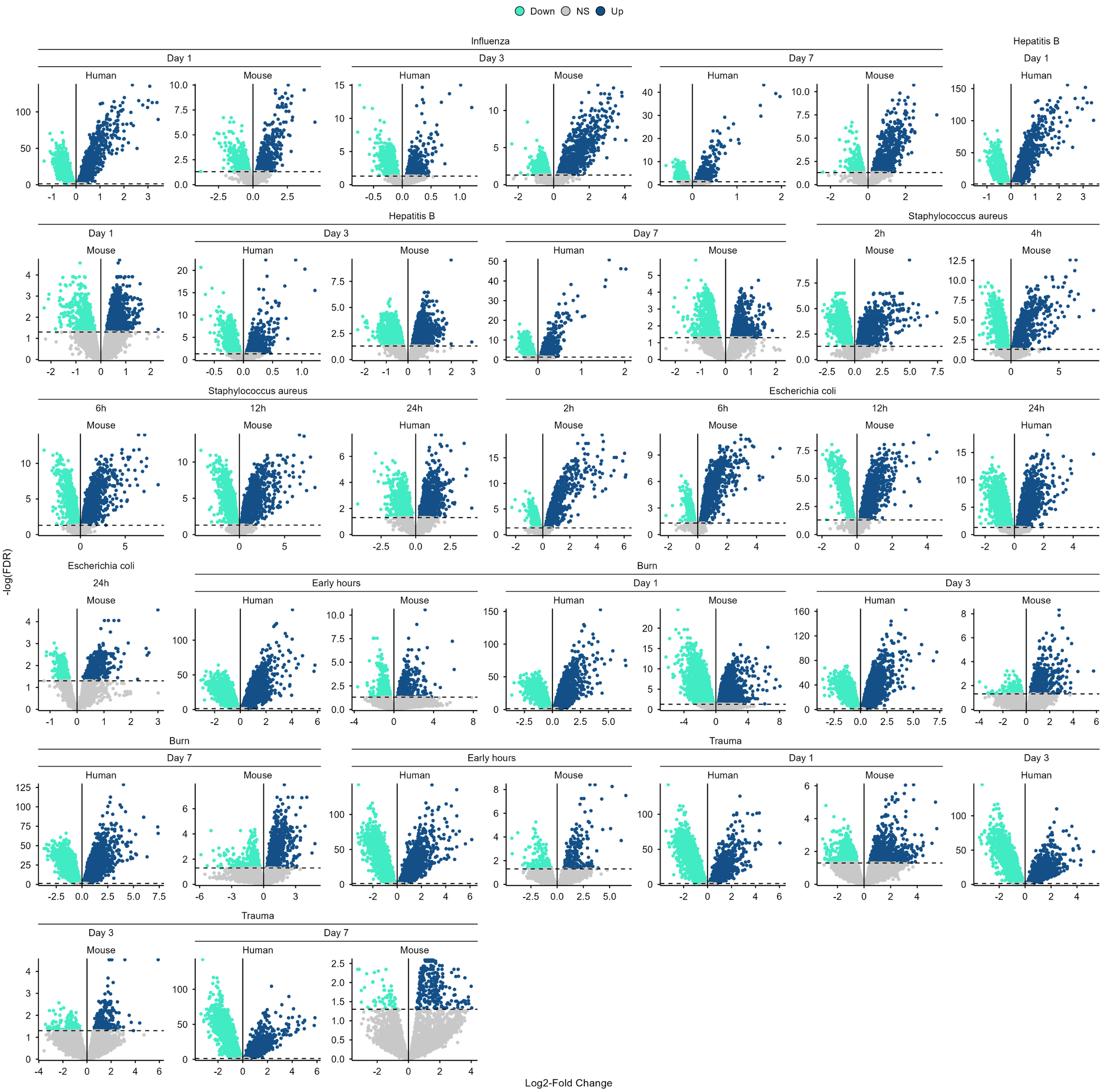
Principal component analysis (PCA) of gene expression profiles from mice and humans under several conditions. The PCAs were generated by using (a) log2-expression and (b) log2-fold changes. Days are represented by shades of blue, and species by shapes (humans = circles; mice = triangles).

Figure S7





Assessment of p-value and adjusted p-value distributions, before and after differential gene expression analysis. Histograms display the distribution of raw (a) and adjusted (b) p-values across all study conditions and timepoints for both human and mouse datasets.



Volcano plots of vaccination, infection, and injury models. Up-regulated genes are shown in dark blue and down-regulated genes in cyan.